RED SEAL EXAM PREP

310S Automotive Service Technician

Free 50-Question Mock Exam

Practice for the Red Seal / Certificate of Qualification exam

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https://redsealquiz.ca · Edition October 2026

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Mock Exam — answer all questions, then check the key

Q1. **A mass air flow (MAF) sensor reads lower than the airflow calculated from RPM and volumetric efficiency at every engine speed, and the gap between the two grows as airflow increases. Long-term fuel trim is only slightly positive at idle but climbs steeply under highway load. The MOST likely cause is:**

A) Air filter restriction causing low airflow

B) Air entering the intake after the MAF sensor

C) MAF sensor contaminated with oil deposits

D) Throttle body dirty causing incorrect air metering

Q2. **A vehicle pulls consistently to the left on a flat, level road with no crown. The two front tires are swapped side to side and the pull stays on the left. The front tires are then moved to the rear and the pull is still on the left. Both front rotors are the same temperature after the drive and both front wheels spin freely when raised. What caused the original pull?**

A) A defective (conical) tire causing the pull

B) A loose left front wheel bearing with play

C) Caster or camber off at the left front

D) Brake drag at the left front caliper

Q3. **How does R-1234yf differ from R-134a in flammability, and what does that mean for servicing?**

A) Both are non-flammable, so recovery, recharge, and leak-test service procedures are identical for the two refrigerants

B) R-1234yf is A2L (mildly flammable), R-134a is A1 (non-flammable) — use A2L-rated equipment, no open flames

C) R-134a is the mildly flammable refrigerant while R-1234yf is completely non-flammable in all service conditions

D) R-1234yf is A3 (highly flammable like propane) and is therefore prohibited from being serviced inside an enclosed shop

Q4. **A technician is diagnosing a parasitic battery drain. After pulling fuses one by one, the draw drops from 450mA to 35mA when fuse #14 (Body Control Module) is removed. What is the BEST next step?**

A) Install a higher-amperage fuse in position #14

B) Unplug components on that fuse circuit one at a time

C) Replace the Body Control Module without further testing

D) Clear all BCM codes and retest after a full drive cycle

Q5. **A vehicle has a hard start condition — the engine cranks normally but takes 3–4 seconds to start. Fuel pressure bleeds down to zero within five minutes of key-off. The most likely cause is:**

A) A faulty crankshaft position sensor delaying start-up timing

B) A clogged fuel filter restricting flow to the injectors

C) A weak battery causing slow fuel pump operation

D) A leaking injector or failed fuel pump check valve

Q6. **An engine overheats only in stop-and-go city traffic and runs at normal temperature on the highway. The coolant level is correct, the coolant is clean, and the thermostat has been verified to open at the correct temperature. What does this pattern point to?**

A) A thermostat stuck closed, restricting coolant flow to the radiator

B) The cooling fan or its control circuit is not moving enough air

C) A head gasket leak pushing combustion gas into the cooling system

D) A worn water pump impeller unable to circulate coolant at any speed

Q7. **An upstream oxygen sensor sits at a fixed high (rich) voltage with no switching. Long-term fuel trim on that bank is near zero, and the signal does not move when the technician deliberately drives the mixture rich and then lean. The MOST likely cause is:**

A) Fuel rail pressure that is far below specification

B) A contaminated or failed sensor stuck reading rich

C) A catalytic converter working at peak efficiency now

D) An engine running too lean at all speeds and loads

Q8. **A dual-zone system blows cold air on the driver side and warm air on the passenger side regardless of the temperature setting. What is the MOST likely cause?**

A) A low refrigerant charge that reduces cooling capacity across the system

B) A failed blend door actuator or a broken blend door on the affected zone

C) A clogged cabin air filter restricting total airflow to both zones

D) A failed compressor clutch preventing the compressor from engaging

Q9. **A high-mileage engine is pushing oil past the rear main seal, and the dipstick lifts out of its tube while the engine idles. The oil level is correct, the engine does not overheat, and there is no coolant loss. The MOST likely cause is:**

A) A restricted PCV system that cannot vent crankcase blow-by

B) Worn rings — crankcase pressure always means ring wear

C) A failed oil pump forcing oil past the seals under pressure

D) Engine oil that is too thin, letting it escape past seals

Q10. **A brake-by-wire (BBW) system on a hybrid vehicle has no traditional brake booster. How is brake assist force generated?**

A) An electric motor-driven hydraulic pump builds pressure on demand

B) The driver simply applies more pedal force by design

C) The drive motor reverses to provide all of the braking force

D) An electric vacuum pump supplies conventional vacuum assist

Q11. **During a brake fluid service, a technician finds that the brake fluid boiling point is significantly lower than specification. What is the MOST LIKELY cause and what is the risk?**

A) Wrong viscosity grade fluid with a lower boiling point

B) Contamination with engine oil from a master cylinder leak

C) Oxidation thickening the fluid and lowering its boiling point

D) Moisture absorbed by the hygroscopic fluid over time

Q12. **What brake fluid specification is required for most vehicles equipped with ABS and electronic stability control?**

A) Glycol-based DOT 3, DOT 4, or DOT 5.1, as the OEM specifies

B) Mineral oil — ABS modulators require non-hygroscopic fluid

C) DOT 5 (silicone-based) — required for all ABS-equipped vehicles

D) Power steering fluid, since both are sealed hydraulic systems

Q13. **A technician diagnoses a P0894 code: "Transmission Component Slipping." Line pressure tests are within specification. What area should be investigated next?**

A) The oil pump output, since correct pressure eliminates the pump

B) Clutch pack clearance, worn friction plates, or a weak solenoid

C) The flexplate/flywheel, as it is the most common cause of P0894

D) The engine ignition timing, as retarded timing overloads the unit

Q14. **An 8-speed automatic transmission is displaying harsh shifts and a fault code for "Transmission Control Module — solenoid circuit fault, shift solenoid E." What is the likely consequence if this solenoid fails permanently open?**

A) The engine torque is limited to idle when the fault is detected

B) The transmission enters limp-home mode, locked in one gear

C) The transmission locks out all forward gears — reverse only

D) Only the commanded gear is affected — other gears work normally

Q15. **What does a MAP sensor measure, and how does that reading affect fuel injection?**

A) Oil pressure — engine bearing protection

B) Intake manifold pressure — engine load

C) Fuel rail pressure — injector opening

D) Exhaust back pressure — ignition timing

Q16. **What is the purpose of a shunt resistor in an automotive electrical system?**

A) To bypass a failed component and keep the circuit working

B) To add resistance in high-current circuits to prevent damage

C) To measure current from the voltage drop across it

D) To limit the current flowing to sensitive electronics

Q17. **During a wheel alignment the rear thrust angle exceeds the manufacturer's specification, and rear toe is not adjustable on this vehicle. What is the correct procedure?**

A) Repair the rear suspension, then set front toe to the thrust line

B) Set front camber to offset the thrust angle and re-check tracking

C) Record the reading only — thrust angle does not affect tracking

D) Adjust front caster until the vehicle stops pulling to one side

Q18. **What is brake fade and what are the two main types?**

A) Gradual loss of braking efficiency; types are lining fade (overheated friction material) and fluid fade (boiling brake fluid)

B) Gradual loss of braking efficiency; types are mechanical fade (worn pads and scored rotors) and hydraulic fade (low reservoir fluid level)

C) Brake noise from glazed friction material; types are squeal fade (high-frequency pad vibration) and groan fade (low-speed shudder when stopping)

D) Increased pedal travel caused by air in the lines; types are static fade (parked on a grade) and dynamic fade (only while the vehicle is moving)

Q19. **What is the purpose of ABS (Anti-lock Braking System) and how does it prevent wheel lockup?**

A) ABS raises pressure at all wheels at once to shorten the stop

B) ABS applies the parking brake at the rear to stabilize the vehicle

C) ABS disables the brakes briefly to let the tires regain traction

D) ABS modulates pressure at each wheel to prevent wheel lockup

Q20. **A customer complains of a rapid ticking from the top of a port-injected engine, new since the last service and getting louder. The noise increases with RPM. Oil level and oil pressure are normal. The MOST likely cause is:**

A) Normal injector noise from the fuel rail

B) Excessive valve clearance or a stuck lifter

C) Crankshaft main bearing knock under load

D) An exhaust manifold heat shield rattling

Q21. **A transmission slips in 2nd and 4th gear but operates normally in 1st, 3rd, and reverse. Using your knowledge of planetary gear sets, what component is MOST LIKELY faulty?**

A) The valve body main pressure regulator affecting all gears

B) The output shaft bearing allowing the shaft to move axially

C) The forward clutch pack that is common to all forward gears

D) The clutch pack or band applied only in 2nd and 4th gear

Q22. **A vehicle has excessive negative camber (top of tire tilts inward) on the front right wheel after a curb strike. What is the most likely damaged component?**

A) A worn wheel bearing with excessive play

B) A broken anti-roll bar link allowing suspension droop

C) A collapsed coil spring on the affected corner

D) A bent strut, lower control arm, or steering knuckle

Q23. **A vehicle wanders on the highway and requires constant steering correction. Which component should be checked FIRST?**

A) Engine and transmission mounts

B) Front wheel hub bearings

C) Front brake caliper slides

D) Tire pressure and tire wear

Q24. **When diagnosing a CAN bus communication fault (U-codes), what does a "Bus Off" condition indicate?**

A) A module counted too many errors and left the bus

B) Outdated scan tool software losing communication

C) The vehicle's power mode is in the accessory-off state

D) The CAN wiring has been disconnected from the battery

Q25. **A vehicle with Electronic Stability Control (ESC) applies the brake at one wheel during normal cornering. No warning lights are on, and a scan tool shows the steering angle sensor at zero offset when the vehicle is driven straight ahead. What is the MOST LIKELY cause?**

A) Low brake fluid causing proportioning valve malfunction

B) A faulty wheel speed sensor reporting wrong speed

C) A sticking ABS solenoid in the hydraulic control unit

D) A brake light switch that is out of adjustment

Q26. **Why must A/C refrigerant be recovered during service rather than released to the atmosphere?**

A) Venting the old charge first lets the pump pull a deeper vacuum, so recovery is only needed if the system still holds pressure

B) R-1234yf has a very low global warming potential, so small amounts may be vented in a ventilated bay

C) Venting refrigerant is illegal and environmentally harmful — it must be recovered with a recovery machine

D) Recovery equipment is required only for ozone-depleting R-12; R-134a and R-1234yf may be discharged through the service port

Q27. **A technician performs a relative compression test using a scan tool while cranking. Cylinder 4 shows significantly higher cranking RPM than the others during its compression stroke. This indicates:**

A) Cylinder 4 has LOW compression

B) Cylinder 4 injector is delivering more fuel

C) The crankshaft is bent or twisted

D) Cylinder 4 has higher compression than others

Q28. **A customer complains of a clunking noise from the front suspension when going over bumps. The MOST likely component to check first is:**

A) Power steering fluid level

B) Front wheel alignment settings

C) Sway bar links and strut mounts

D) Tire pressure on all four corners

Q29. **A manifold absolute pressure (MAP) sensor reads consistently HIGHER than actual barometric pressure at key-on, engine off. This MOST likely indicates:**

A) A throttle body sticking closed at idle

B) A vacuum leak downstream of the throttle

C) A failed MAP sensor or its signal circuit

D) A plugged or kinked MAP sensor vacuum line

Q30. **A stall test is performed to the specified procedure on an automatic transmission: brakes firmly applied, a drive range selected, throttle held wide open only for the few seconds allowed. The engine speed reached is well ABOVE the specified stall speed. What does that indicate?**

A) The engine is not producing its full power output

B) The converter stator one-way clutch is not holding

C) The torque converter clutch is applying too early

D) A clutch pack or band inside the unit is slipping

Q31. **What is the purpose of transmission fluid in an automatic transmission?**

A) It acts only as a coolant for the torque converter assembly

B) It lubricates, applies clutches, cools, and fills the converter

C) It only lubricates the moving parts to prevent internal wear

D) It provides only the hydraulic pressure needed for gear shifts

Q32. **In an A/C system that uses a thermostatic expansion valve (TXV), which component stores liquid refrigerant and desiccant, and where is it located?**

A) A muffler on the compressor discharge line

B) A receiver-drier on the high (liquid) side

C) An accumulator on the low (suction) side

D) The evaporator core on the low side

Q33. **A scan tool shows an active code P0128 — Coolant Temperature Below Thermostat Regulating Temperature. What is the MOST likely cause?**

A) The coolant temperature sensor has failed high (reads too cold)

B) Low coolant level causing the sensor to read incorrectly

C) The radiator fan is running continuously at full speed

D) A stuck-open thermostat keeping the engine too cool

Q34. **A front-wheel-drive vehicle makes a clunk from the front suspension only when it is driven over a bump while turning under power, such as pulling over the lip of a driveway. With the vehicle standing still, turning the steering lock to lock produces no noise, and steering effort and return to centre are normal. The most likely cause is:**

A) A loose upper strut mount bearing

B) Air ingestion in the rack from low power steering fluid

C) A worn shock absorber topping out

D) A worn outer CV joint with play

Q35. **An engine has a stretched timing chain. How would this most likely affect cam timing and what symptom would it produce?**

A) No effect on timing — the hydraulic tensioner takes up the slack and fully compensates for chain stretch

B) Cam timing becomes erratic — the only symptom is a ticking noise at idle

C) Cam timing advances — causes high idle, surging, and spark knock under load

D) Cam timing retards — causes rough idle, low power, and may set VVT fault codes

Q36. **A dual-clutch transmission (DCT) hesitates and jerks during slow-speed manoeuvring in parking lots and traffic, and is normal at highway speeds. The owner reports it did not do this when the vehicle was new and that it has worsened steadily over the last two years. Fluid level and condition are correct and there are no leaks. What is the MOST LIKELY cause?**

A) Contaminated fluid stopping DCT operation at any speed

B) A failed speed sensor mis-selecting gears at low speed

C) Clutch engagement adaptation values drifting out of range

D) An inability of DCTs to operate at low speeds by design

Q37. **What does the term "firing order" refer to in a multi-cylinder engine?**

A) The sequence in which cylinders receive fuel injection

B) The sequence of valve timing events

C) The order spark plugs are replaced during a tune-up

D) The sequence in which cylinders produce a power stroke

Q38. **A circuit has a 12V battery, a 4-ohm resistor, and a 2-ohm resistor connected in series. What is the total current flow?**

A) 6 amps

B) 4 amps

C) 3 amps

D) 2 amps

Q39. **A vehicle keeps bouncing two or three times after one end is pushed down and released, and the front tires show cupped, scalloped patches around the tread. What is the MOST LIKELY cause?**

A) Front wheel alignment toe set outside specification

B) Worn shock absorbers no longer damping the suspension

C) Front wheels out of balance, felt at highway speed

D) Sagging coil springs that have lost their free height

Q40. **A vehicle's EVAP system leak detection test stores a small leak code (P0442). The FIRST item to check is:**

A) Replace the EVAP purge control valve

B) Inspect and tighten the fuel filler cap

C) Smoke test the charcoal canister vent hose

D) Replace the charcoal canister assembly

Q41. **An ABS system stores a right rear wheel speed sensor fault code. The sensor resistance is within spec and voltage is present. The vehicle still activates ABS on every stop. The NEXT diagnostic step is:**

A) Check the brake pad thickness on that corner

B) Replace the wheel bearing assembly

C) Replace the ABS hydraulic control module

D) Inspect the tone ring for damaged or caked teeth

Q42. **With the engine off, a technician presses the brake pedal several times to use up the reserve vacuum, then holds the pedal down and starts the engine. On a vehicle with a sound vacuum booster and check valve, what should happen?**

A) The pedal should sink slightly under the foot

B) The pedal should stay exactly where it is

C) The pedal should pulse in time with the engine

D) The pedal should rise firmly against the foot

Q43. **An engine calls for a 5W-30 engine oil. What does the W in that grade designation mean?**

A) It stands for weight: the two numbers are cold and hot weights

B) It stands for wear: the first number rates the anti-wear additive

C) It marks the oil as a synthetic blend rather than conventional

D) It stands for winter: the first number rates flow when cold

Q44. **A dual-clutch transmission (DCT) hesitates and jerks slightly during low-speed manoeuvring in parking lots. The vehicle has behaved this way since it was new, no transmission codes are stored, the clutch adaptation values read within range, and a same-model vehicle on the lot behaves identically. This is BEST described as:**

A) A serious mechanical failure needing immediate repair

B) A software fault in the TCM that no update can correct

C) A known low-speed characteristic of DCT clutch control

D) Proof that both clutch packs are worn out and need replacing

Q45. **When diagnosing automatic transmission shift quality, what does "hunting" (frequent up-down shifting at a steady speed) indicate?**

A) A worn torque converter causing inefficient power transfer

B) A shift point calibration, TPS, or shift solenoid fault

C) A low fluid level preventing full clutch pack engagement

D) Worn friction plates slipping in the direct clutch pack

Q46. **A scan tool shows an upstream oxygen sensor (bank 1 sensor 1) reading stuck rich at a consistently high voltage. Long-term fuel trim on that bank is strongly negative, and the sensor still responds when the mixture is forced lean. What is the MOST likely cause?**

A) A vacuum leak downstream is causing a false rich indication

B) The catalytic converter has failed, creating a false rich reading

C) The oxygen sensor has failed internally — replace the sensor

D) The engine really is rich — check injectors, pressure, and MAF

Q47. **What is "brake pedal pulsation" and what is the most common cause?**

A) Excessive pedal travel caused by badly worn brake pads

B) A soft pedal that gradually firms — caused by air in the lines

C) A braking vibration in the pedal from rotor thickness variation

D) Intermittent brake grabbing from a sticking caliper piston

Q48. **A vehicle with rear drum brakes has excessive brake pedal travel. The fluid level is correct, the pedal is firm once it comes up, it does not sink under steady pressure, and bleeding produces no air. What is the MOST likely cause?**

A) The rear wheel cylinders are leaking past the cups

B) The master cylinder is bypassing internally

C) The front brake pads are worn near the limit

D) The rear shoe adjusters have stopped working

Q49. **An automatic transmission fitted with a dipstick is being checked for fluid level. The vehicle is level and the selector is in the position the service information specifies. Which further condition gives a correct reading?**

A) Engine switched off and the vehicle left overnight to drain back

B) Engine switched off, transmission cold, vehicle on a hoist

C) Engine running at fast idle, transmission still cold from the start

D) Engine running at idle, transmission at operating temperature

Q50. **A port-injected engine loses power only under sustained high load, such as towing or a long climb, and runs normally otherwise. Fuel pressure measured at key-on and at idle is within specification, and the injectors are electrically sound. Which test is MOST likely to identify the fault?**

A) A cylinder balance test at idle in the shop with no road load

B) A fuel volume (delivery rate) test with the engine under load

C) A repeat static pressure test with the key on, engine not running

D) A resistance check of each injector winding, connectors removed

Answer Key & Explanations

> Scoring: 35 correct out of 50 is 70%. These 50 questions come from the 139-question 310S bank; the last page shows what the complete printable bank contains.

Q1. Answer: C

Read the trim pattern, not just the low reading. A MAF element coated with oil — typically from an over-oiled reusable filter — or with fine dust is insulated, so it gives up its heat more slowly and under-reports airflow. The error scales with airflow: small at idle, large at load. The ECM fuels from that low reading, so the engine leans out most where the error is biggest, which is exactly the trim pattern described. Unmetered air entering after the sensor produces the opposite pattern, because the leak is a large fraction of total airflow at idle and a small fraction once the throttle opens — those vehicles show their worst trims at idle and clean up under load. A restricted air filter genuinely reduces airflow, and the MAF reports that reduced airflow honestly, so fuel still matches air and trims stay near zero. A dirty throttle body upsets idle air control and idle quality, not MAF accuracy. Compare grams per second at idle and at wide-open throttle against expected values, then clean the element with MAF-specific cleaner and recheck before condemning the sensor.

Q2. Answer: C

A pull that survives both moves is not coming from a tire — it belongs to the corner. Conicity and plysteer are built-in lateral forces: the tire pushes sideways in a fixed direction relative to itself, so moving it from one front position to the other inverts the direction of that push. The pull then reverses, or cancels against the other front tire, and once the tire is at the rear its lateral force has little effect on steering. Neither happened here, so the tires are ruled out. Equal rotor temperatures after the drive and wheels that spin freely rule out brake drag at the left front, and a loose wheel bearing shows as play at the wheel, noise and wander rather than a steady one-way pull. What is left is the alignment angles that generate side force at a steered wheel: the vehicle leads toward the side with the more positive camber and toward the side with the less positive caster, so the left front is the corner to measure. Put the vehicle on an alignment rack, check ride height first because a sagged spring moves both angles, then compare left and right caster and camber before any part is replaced. Radial force variation is a separate defect: it varies the tire's radial stiffness through each revolution and shows as vibration and shake, not as a pull.

Q3. Answer: B

R-1234yf is classified A2L (mildly flammable); R-134a is A1 (non-flammable). A2L is the lower-flammability class. R-1234yf burns slowly — a burn velocity of about 1.5 cm/s against 46 cm/s for propane — and its heat of combustion is roughly a quarter of propane's. It also takes thousands of millijoules to ignite, against about 0.25 mJ for propane, so it is far harder to set alight than a hydrocarbon refrigerant. It will still burn, so servicing requires A2L-rated recovery/recharge equipment, adequate ventilation, a refrigerant-specific electronic leak detector, and no open flames or ignition sources near the work. Systems are also designed with features to manage the small flammability risk.

Q4. Answer: B

Isolating a parasitic draw: after identifying the circuit, systematically unplug components on that circuit. The BCM fuse powers multiple subsystems. Use a wiring diagram to identify all circuits on that fuse, then unplug each module/component on the circuit one at a time while monitoring current draw to isolate the specific drain source. Many BCM-related drains are caused by accessories that the BCM controls — door modules, lighting modules, etc. — staying awake when they shouldn't. A wiring diagram is essential to find all components.

Q5. Answer: D

Pressure bleeds off after key-off, then a long crank: check valve or injector leak-down. A healthy system holds residual rail pressure long after shut-off. Manufacturers state the limit as a pressure retained within a small tolerance over a stated timed interval, so compare the reading against the service information for that vehicle rather than against a rule of thumb. When pressure bleeds back, either the check valve in the fuel pump is letting fuel drain to the tank (replace the pump) or an injector is dripping into a cylinder. The extra cranking time is the pump refilling and re-pressurising the rail before the engine can fire. A restricted filter would show as low pressure with the engine running, and a weak battery would show as slow cranking — neither is present here.

Q6. Answer: B

The driving condition that produces the complaint tells you which side of the system to look at. At road speed, ram air forced through the radiator does most of the cooling and the fan contributes very little. In stop-and-go traffic there is no ram air, so the fan is the only source of airflow through the core. A complaint that appears only at low vehicle speed and clears on the highway therefore points at airflow, not at coolant flow: a fan that does not run, a failed relay or control module, a fan clutch that will not lock up, a shroud left off or broken, or a core packed with debris. Faults on the coolant side behave the opposite way — a stuck-closed thermostat, a failed pump, or combustion gas in the coolant all add heat or reduce flow, so they get worse as speed and load rise and do not clear on the highway. Verify by commanding the fan with a scan tool and confirming it actually turns, then inspect the core, the shroud and the fan clutch before condemning anything expensive.

Q7. Answer: B

A sensor that will not respond to a forced mixture change has failed — the mixture is not the problem. A healthy upstream zirconia sensor switches rapidly between a low (lean) and a high (rich) voltage as the ECM cycles the mixture. Two very different faults can park that signal high, and the fuel trims separate them. If the engine really were rich, the ECM would be subtracting fuel and long-term fuel trim would sit well negative. Here the trim is near zero and the signal does not move when the mixture is deliberately driven rich and then lean, so the sensor is not reporting what the engine is doing. Contamination (silicone, coolant, oil ash), an internal failure, or a heater circuit fault that leaves the element cold will all produce a dead, fixed signal. Confirm on a scope before replacing, and check the heater circuit and the sensor ground before condemning the sensor itself.

Q8. Answer: B

A blend door directs air through or around the heater core to set temperature; a failed actuator or broken door on one zone loses temperature control for that zone. Low charge or a failed compressor would affect cooling on both sides, not one zone. A failed blend door actuator often produces a clicking noise and sets an HVAC actuator fault code. Confirm actuator movement with a scan tool bidirectional test.

Q9. Answer: A

Oil pushed out of seals and a dipstick that lifts are symptoms of crankcase pressure, and the crankcase ventilation path is what is supposed to relieve it. The PCV system draws blow-by out of the crankcase using intake vacuum and replaces it with filtered air, holding the crankcase at or slightly below atmospheric pressure. Plug the PCV valve, its hose, or the oil separator, and the blow-by has nowhere to go: pressure rises and forces oil out wherever the seal is weakest — the rear main, the valve cover gaskets, the dipstick tube. The leak is the symptom; the restriction is the cause. Prove the ventilation path clear first, checking the valve, hoses and separator, and confirm with a low-range pressure gauge or manometer at the dipstick tube. Worn rings do raise crankcase pressure too, but that is established by a cylinder leakage or blow-by test after ventilation has been ruled out — it is not the automatic conclusion. Oil pump pressure acts inside the galleries, not on the crankcase, and replacing seals while the crankcase is still pressurized simply repeats the failure.

Q10. Answer: A

Brake-by-wire: electric hydraulic pump provides assist, blends with regen braking. Hybrid/EV brake systems use electric-motor-driven pumps (or electrohydraulic actuators) to provide brake pressure on demand without a conventional vacuum booster. The system must seamlessly blend regenerative braking (electric motor generates electricity while slowing) with friction brakes — prioritizing regen for energy recovery and using friction brakes to supplement, managing the transition smoothly. This requires sophisticated control.

Q11. Answer: D

DOT brake fluid is hygroscopic — absorbs moisture from the atmosphere through brake system seals and hoses. Water contamination significantly reduces the wet boiling point of brake fluid and increases the risk of vapour lock during heavy braking. DOT 3: dry 205°C / wet 140°C. DOT 4: dry 230°C / wet 155°C. Moisture-laden fluid can boil during sustained heavy braking (mountain descents, track use), creating vapour bubbles that are compressible — resulting in a spongy or non-existent brake pedal (vapour lock). This is why brake fluid should be replaced every 2–3 years regardless of appearance.

Q12. Answer: A

ABS/ESC vehicles: use the OEM-specified DOT rating, usually DOT 4 or DOT 5.1. DOT 3, DOT 4, DOT 4 LV and DOT 5.1 are all glycol-based and miscible with one another, so the system is filled and topped up with the grade the manufacturer calls out. DOT 5 is silicone-based and is not miscible with the glycol fluid these systems are filled with; because it does not absorb moisture, free water collects at low points where it corrodes components and can boil under hard braking, and silicone fluid aerates and compresses more readily under rapid ABS modulator cycling. ABS and ESC vehicles are commonly specified DOT 4, DOT 4 LV or DOT 5.1 for both a higher wet boiling point and a lower cold viscosity, so the modulator pump can move fluid quickly in winter. Never mix DOT 5 with DOT 3, DOT 4 or DOT 5.1, and confirm the specification on the reservoir cap or in the service information before adding fluid.

Q13. Answer: B

P0894 (slipping) with normal line pressure: focus on clutch pack wear, clearance, or solenoid control of specific elements. Normal line pressure confirms the pump and main pressure regulator are working. Slipping must be occurring at a specific clutch pack or band. Causes: worn friction plates (excess clearance), weak apply pressure at a specific solenoid circuit (solenoid not fully applying that clutch), or worn clutch pack seals. Use a clutch application chart to identify which elements are in use during slipping conditions.

Q14. Answer: B

Shift solenoid fault triggers limp-home mode — transmission locks into a safe fixed gear. Modern transmissions with electrical faults default to limp-home (fail-safe) mode. The TCM locks the transmission in one gear (commonly 2nd or 3rd) to allow the vehicle to be driven safely to a service facility while preventing further damage. This prevents: multiple gear engagement simultaneously (mechanical damage), loss of all drive. The driver experiences loss of all shifting ability, engine RPM limited by the fixed gear, and difficulty accelerating from stop.

Q15. Answer: B

MAP = engine load sensor. It measures absolute pressure in the intake manifold. High vacuum (low absolute pressure) means light load with little air entering, so the ECM commands a shorter injector pulse width — less fuel, not a leaner mixture. Low vacuum (high absolute pressure) means heavy load with more air entering, so pulse width increases. The ECM combines MAP with engine speed and intake air temperature to work out the mass of air entering the cylinders, then meters fuel to the target ratio. Oil pressure, fuel rail pressure and exhaust back pressure each have their own dedicated sensor.

Q16. Answer: C

A shunt is a low-value, high-precision resistor placed in series so current can be worked out from the voltage drop across it. Measuring the millivolt drop across a known, precise low resistance gives the current directly (I = V/R). Vehicles use this to monitor charge and discharge current: the battery current sensor at the negative battery terminal carries the whole ground-path current through a very low-value shunt and reports it to the module that manages state of charge. A clamp-type ammeter is not a shunt device - it senses the magnetic field around the conductor, which is why it can read current without opening the circuit. The shunt has to be in series, but its resistance is low enough that it does not meaningfully change how the circuit behaves.

Q17. Answer: A

A rear axle out of thrust specification is a repair job, not a front-end adjustment. Thrust angle is the direction the rear axle points relative to the vehicle centreline, and it is set by rear toe. Out of specification, the vehicle dog-tracks and the steering wheel sits off centre. Nothing at the front corrects it: camber and caster do not change where the rear axle points, and dialling front toe alone only straightens the steering wheel while the vehicle still crabs down the road. With rear toe non-adjustable, inspect the rear end for a bent axle or trailing arm, a shifted cradle or subframe, collapsed or displaced bushings and collision damage, then reposition, shim or replace whatever is out of position — many non-adjustable rear suspensions accept offset bushings or shims for exactly this. Once the rear is back within specification, set front toe to the thrust line rather than to the geometric centreline.

Q18. Answer: A

Brake fade: reduced braking effectiveness from heat. Two types: lining fade and fluid/vapor lock fade. Lining fade: friction material overheats, its coefficient of friction drops, reducing braking force despite normal pedal pressure. Fluid fade (vapor lock): brake fluid boils, creating compressible gas bubbles — pedal goes low and spongy. Both occur during extended/severe braking (long downhill grades). Solutions: upgraded brake fluid (higher dry/wet boiling point), cross-drilled/slotted rotors, performance brake pads.

Q19. Answer: D

ABS modulates brake pressure per wheel to maintain tire slip in optimal range (10–20%) for maximum braking with steering control. Wheel speed sensors detect impending lockup (sudden deceleration). The ABS modulator releases, holds, then reapplies brake pressure in rapid cycles (10–15 times/sec). A locked tire has less friction than a rolling tire at controlled slip. ABS allows steering while braking hard — critical for obstacle avoidance.

Q20. Answer: B

Valve train tick: top of the engine, RPM-proportional, oil pressure normal. Excessive valve clearance or a collapsed or stuck hydraulic lifter gives a light, regular tick that follows engine speed. Injector tick is present on every running engine from the day it was built and does not appear between services or grow louder; it is loudest on direct-injection engines, where the high-pressure injectors and pump are genuinely noisy by design. A main bearing knock is heavier and lower in the engine and worsens under load rather than with bare RPM. A heat shield buzzes or rattles at particular engine speeds rather than ticking steadily. Check valve clearance and hydraulic lifter condition. Some tick at cold start-up is normal until the lifters prime with oil.

Q21. Answer: D

Slipping in specific gears indicates the component common to those gears has failed. Automatic transmission planetary systems use a different combination of clutch packs and bands for each gear. Many designs name a holding element for the gears it applies in, so the element applied in second and fourth is commonly labelled the 2-4 band or the 2-4 clutch; failure of that one element causes slipping in those two gears while the others are unaffected. This approach — identifying which gears fail, then tracing to the common holding element on the manufacturer's clutch and band application chart — is key to transmission diagnosis without complete disassembly.

Q22. Answer: D

Camber beyond adjustment range after impact: structural damage. Camber is primarily a function of suspension geometry — strut angle, control arm length, and knuckle design. These structural components control the camber angle; a hard impact can deform them, changing camber beyond the adjustment range permanently. Inspect with a straightedge and alignment rack. Replace bent components before alignment.

Q23. Answer: D

Highway wander: start simple. Underinflated tires or uneven tire wear cause wandering, so check them first. Then move to the steering linkage and alignment: tie rods, steering gear/rack play, and front wheel alignment (especially caster angle — low caster = wandering, directional instability).

Q24. Answer: A

"Bus Off" means a CAN module has counted too many transmit errors and taken itself off the network so it cannot disrupt the other modules. Error confinement is built into the protocol: each node keeps a transmit and a receive error counter. A node becomes error-passive once its transmit error counter passes 127, and it goes bus-off once that counter exceeds 255, at which point it stops transmitting altogether. Causes: shorts in the CAN wiring, a failing module, or severe electromagnetic interference. The remaining modules may still communicate, and the missing data from the silent module sets U-codes across the network. Find the root cause in the wiring or the module rather than clearing the codes and handing the vehicle back.

Q25. Answer: B

A wheel speed sensor that reports a plausible but wrong speed makes the ESC module correct a slide that is not happening. ESC compares the four wheel speeds against steering angle, yaw rate and lateral acceleration to judge whether the vehicle is following the path the driver asked for. A sensor with erratic output can make the module read one wheel as spinning or locking and brake it, and because the signal stays inside the plausible range nothing is stored and no lamp comes on. The other classic cause of single-wheel intervention is a steering angle sensor zero-point offset, which is why the scan tool check driving straight matters here - that input has already been cleared. A brake light switch out of adjustment only tells the module the driver is on the brake; it cannot command an apply. A sticking hydraulic solenoid can hold or release pressure the driver has already applied but cannot generate one, and low fluid does not produce single-wheel braking. Diagnose with live data, comparing all four wheel speeds with vehicle speed.

Q26. Answer: C

Venting refrigerant is prohibited by law and environmentally harmful; it must be recovered and recycled with a proper recovery machine. In Canada, refrigerant handling is regulated (federal and provincial environmental regulations) and technicians require refrigerant-handling certification. Recover the charge before opening the system, use refrigerant-specific equipment, and never mix refrigerants in a recovery machine.

Q27. Answer: A

Higher cranking speed during a compression stroke = LOW compression in that cylinder. The starter's load drops when a cylinder has little compression to build, so with less resistance the crankshaft briefly speeds up. This is the scan tool form of the relative compression test: the tool graphs crankshaft speed taken from the crankshaft position sensor, and no gauge is fitted. The lab scope form of the same test watches starter current instead, where the weak cylinder shows the mirror image — a shallow current hump, because it loads the starter less. Both forms point at the same cylinder and neither gives a pressure figure, so confirm with a compression gauge and then a leak-down test.

Q28. Answer: C

Clunking over bumps = loose or worn suspension components. Most common causes: sway bar end links and bushings (common, inexpensive), strut mount bearing (also causes clunk on steering), worn ball joints, loose control arm bushings. Each should be checked while the suspension is unloaded (lift the vehicle) and by pushing/pulling components for play.

Q29. Answer: C

With the engine stopped there is no vacuum anywhere in the system, so no vacuum-related fault can be the answer. Key-on, engine off, the intake manifold sits open to atmosphere through the throttle and the MAP sensor should report barometric pressure — which falls with altitude, so a sea-level figure is not the standard everywhere. A plugged or kinked hose simply traps air at atmospheric pressure, which is the correct reading under this condition; a leak downstream of the throttle likewise changes nothing when no vacuum exists; and a throttle held closed cannot raise manifold pressure above atmosphere. A reading above barometric with the engine off can therefore only come from the sensor or its wiring: a sensor that has drifted high, a shorted or biased signal line, or a corroded connector raising signal voltage. Compare the MAP and BARO parameters on the scan tool at key-on — they should agree within a few kPa — then back-probe the sensor for reference voltage, ground and signal before replacing it. A blocked or disconnected vacuum line does show itself, but only with the engine running, where MAP stays near barometric instead of dropping.

Q30. Answer: D

Stall speed is the point where the engine's torque output and the converter's ability to absorb it balance, so a reading above specification means torque is escaping behind the converter. With the vehicle held still the converter turbine cannot turn, and the engine winds up until the converter is absorbing everything the engine can produce. If a holding element inside the transmission - a clutch pack or a band for the range selected - cannot hold, the driveline gives way before the converter loads the engine fully, so the engine runs past its specified stall speed. A reading BELOW specification means the opposite: either the engine cannot make its rated torque, or the converter's stator one-way clutch is freewheeling instead of holding, so the converter never multiplies torque and the engine reaches its ceiling early. Compare the reading with the figure for that engine and transmission combination, repeat it in each drive range to see which ranges misbehave, and read the result alongside a line pressure test and a road test rather than on its own. The test is hard on the unit: hold it only for the few seconds the procedure allows, watch fluid temperature, and let the transmission cool between attempts.

Q31. Answer: B

ATF does four jobs at once: lubrication, hydraulic pressure for clutch and band application, heat transfer, and fluid coupling in the torque converter. It lubricates planetary gearsets, bearings and bushings. It carries the hydraulic pressure that applies the clutch packs and bands to engage each gear. It picks up heat and carries it to the cooler. It fills the torque converter, where it couples the engine to the transmission and multiplies torque. Using a fluid with the wrong friction modifiers causes shift quality complaints, clutch slip and converter shudder.

Q32. Answer: B

TXV systems use a receiver-drier on the high (liquid) side; orifice-tube systems use an accumulator on the low (suction) side. The receiver-drier stores liquid refrigerant, holds desiccant to absorb moisture, and filters debris. The desiccant becomes saturated once the system is opened, so the receiver-drier (or the accumulator on orifice-tube systems) should be replaced whenever the system is opened for major service.

Q33. Answer: D

P0128: thermostat stuck open = engine never reaches operating temperature. A stuck-open thermostat allows too much coolant to circulate prematurely — the engine stays cold longer and may never reach the ECM's expected operating temperature. Symptoms: heater takes very long to get warm, poor fuel economy (cold-run enrichment runs too long), P0128 code.

Q34. Answer: D

The outer CV joint is the only one of these loaded by steering angle, suspension travel and drive torque at the same moment. As its cage and balls wear, the joint develops lash; put it at a large angle, compress the suspension and apply torque, and it takes up that lash with a clunk — the classic driveway-lip noise on a front-drive car. An upper strut mount bearing can clunk too, and on a MacPherson strut it is a common source of noise when turning; what rules it out here is the second half of the stem. A failing mount bearing makes its noise whenever the strut has to rotate, so it can be reproduced by steering the vehicle while it stands still, and it normally stiffens steering effort or slows the return to centre. Neither happens on this vehicle. A shock topping out clunks in rebound over a bump whether or not the vehicle is turning, and air drawn into the rack gives a groan or buzz that follows steering effort rather than bumps. Confirm by checking the axle for play at full lock and looking for a split boot with grease thrown around the inner wheel well.

Q35. Answer: D

Stretched timing chain retards cam timing, reducing power and causing rough running. Chain stretch effectively lengthens the chain, causing the camshaft to lag behind the crankshaft — retarded cam timing. Symptoms: reduced power especially at low-end, rough idle, rattling noise on start-up (tensioner over-extended), and VVT system codes because the ECM cannot achieve target cam advance. Chain stretch beyond tensioner range = timing jumps possible (catastrophic).

Q36. Answer: C

A low-speed complaint that was absent when the vehicle was new and has grown steadily points at clutch wear moving the engagement point beyond what the TCM can adapt to. A DCT learns the point at which each clutch begins to transmit torque and stores it as an adaptation value. As the friction material wears, the real engagement point moves and the TCM shifts its adaptation to follow. Once the required correction runs past the end of the adaptation table, engagement comes too early or too late and the driver feels hesitation and jerking at crawling speeds, where the clutch is being slipped the most. At highway speed the clutch is fully engaged, so the complaint disappears — which is why the condition it appears under is the strongest clue. Read the adaptation values, then perform an adaptation reset and relearn; if the values run out of range again, the clutch is worn past its service limit. Note the contrast with a DCT that has jerked at low speed since new, with adaptation in range and a same-model vehicle behaving identically — that is the normal characteristic of clutch engagement, not wear.

Q37. Answer: D

Firing order: sequence of power strokes across cylinders. The firing order is engineered to balance engine loads and minimize vibration. A typical inline-4 firing order is 1-3-4-2; a V8 may be 1-8-4-3-6-5-7-2. Incorrect firing order (e.g., swapped ignition wires) causes rough running, backfiring, and poor performance.

Q38. Answer: D

Ohm's Law: I = V/R. Series circuit: total resistance = 4 + 2 = 6 ohms. I = 12V / 6Ω = 2 amps. In series circuits, resistances add. Total current is the same through all components. Voltage drops across each resistor proportionally: V1 = 2A × 4Ω = 8V; V2 = 2A × 2Ω = 4V; 8+4=12V (confirms calculation). Understanding Ohm's Law and series/parallel circuits is fundamental to automotive electrical diagnosis.

Q39. Answer: B

A suspension that keeps oscillating has lost its damping, and cupped tire wear is the same fault written on the tread. The spring carries the weight and stores the energy of a bump; the shock absorber or strut converts that energy into heat so the spring stops after one movement instead of oscillating. A worn damper cannot do that, so the corner keeps bouncing after it is pushed and released, and the vehicle floats over dips and feels unsettled at speed. The tire pays for it: as the wheel bounces, the contact patch is alternately loaded and unloaded, and the tread wears into a pattern of high and low patches around its circumference - cupping, or scalloping. The other three answers each show themselves differently. Sagged springs change ride height and therefore the alignment angles, but they do not allow repeated oscillation. Out-of-balance wheels vibrate at a speed tied to wheel rotation and have no effect on the bounce test. Incorrect toe scrubs the tread with a feathered edge across the face of the tread, not in patches around it. Confirm by comparing the bounce at each corner, looking for oil wetting down the damper body, and checking the mounts and bushings before condemning the unit.

Q40. Answer: B

EVAP small leak: check fuel cap first. The fuel cap is the most common source of small EVAP leaks — a damaged seal or cap not fully tightened is responsible for a large percentage of P0442 codes. Inspect the cap O-ring and test with a cap tester. If cap tests good, proceed to smoke test the rest of the EVAP system.

Q41. Answer: D

Correct sensor but wrong signal = tone ring problem. Damaged, cracked, missing, or packed-with-debris tone ring (reluctor ring) teeth create an irregular signal that the ABS module interprets as wheel lockup. This triggers ABS unnecessarily. Clean the tone ring and inspect for damaged teeth. Common on vehicles that have driven through deep mud or water.

Q42. Answer: A

The pedal drops because the booster has just been given the vacuum it needs. A vacuum booster works on the difference between atmospheric pressure on one side of its diaphragm and manifold vacuum on the other, and it uses that difference to add force to whatever the driver applies. Pumping the pedal with the engine off exhausts the reserve vacuum stored in the booster and held there by the one-way check valve, so the last applications feel hard and high - that is the unassisted pedal. Start the engine and vacuum reaches the booster, the diaphragm is pushed, and the assist appears under a foot that is already pressing, so the pedal sinks. A pedal that does not move means the booster is not receiving vacuum or cannot hold it: check the hose for splits or a poor seal, check the one-way check valve, and check the diaphragm. Nothing in this test should push the pedal back up, and a pulsation with the engine running is not part of it. Note what the test does not prove: it checks the assist, not the hydraulics, so a pedal that afterwards sinks slowly toward the floor under steady pressure is a master cylinder bypassing internally and a separate fault.

Q43. Answer: D

A multigrade oil is rated twice, and the W half is the cold rating. The W stands for winter. The number in front of it describes how readily the oil flows and pumps at low temperature: a 0W or 5W oil stays fluid at temperatures where a 15W or 20W oil is thick and slow, so on a cold Canadian morning it reaches the bearings, the camshafts and the chain tensioners sooner and the engine spends less of its life running on the metal-to-metal contact of a dry start. The number after the dash describes the oil's viscosity at operating temperature. Both halves come from the manufacturer's specification and neither can be traded against the other: a heavier cold grade delays oil delivery at start-up, and a hot grade thinner than specified reduces the oil film the engine was designed around. The W is not an abbreviation for weight, and neither number describes an additive level or whether the oil is conventional, blended or full synthetic - those are stated separately on the container, along with the service classification the engine requires.

Q44. Answer: C

Present since new, no codes, adaptation values in range, and a same-model vehicle behaves the same — that is the design characteristic, not a fault. A dual-clutch transmission engages a mechanical clutch rather than a fluid coupling. A torque converter absorbs the speed mismatch between engine and driveline through fluid slip; a DCT has to slip a friction clutch under electronic control instead, and at crawling speeds that control is at its least smooth, particularly on dry-clutch designs. Establish the baseline before condemning anything: check for manufacturer bulletins and TCM calibration updates, read the clutch adaptation values, and road-test a known-good vehicle of the same model for comparison. Note the contrast with a DCT whose low-speed behaviour appeared after the vehicle was new and has worsened over time with adaptation values drifting out of range — that one is a genuine wear fault. History and adaptation data, not the symptom on its own, tell you which case is in front of you.

Q45. Answer: B

Shift hunting: the transmission cycles between two gears at a steady speed — a shift control problem, not a mechanical wear problem. Incorrect shift point calibration, a throttle position sensor fault, or a shift solenoid fault are the usual causes. The TCM uses vehicle speed, throttle position, engine load and gear ratio to decide shift points. If the throttle position signal is erratic, the TCM sees load conditions that keep alternating and commands one shift after another; an erratic vehicle speed signal does the same thing for the same reason, because both are shift-scheduling inputs. Worn friction material, by contrast, shows up as slipping or flare during a shift, not as repeated cycling between two gears. Check: throttle position voltage sweep (smooth?), the vehicle speed signal on a graphing meter, shift solenoid response, and fluid level and condition.

Q46. Answer: D

A rich reading, strongly negative fuel trim, and a sensor that still responds to a forced mixture change together point to a genuinely rich engine. Negative trim on its own does not prove it. A sensor biased high reports rich when the mixture is actually correct, the ECM subtracts fuel to match, and the trim looks exactly the same — and that fault frequently sets no code at all, because the correction stays inside the ECM's acceptance window. What separates the two here is the sensor's behaviour: force the mixture lean and a good sensor swings down to a proper low, lean voltage, while a sensor that cannot follow a commanded change is itself the fault. Confirm with an independent measurement if there is any doubt — a wideband sensor or an exhaust gas analyzer, plus spark plug condition. Then work the rich causes in order of likelihood: an injector leaking or dripping when commanded off (injector balance test), fuel pressure above specification from a failed regulator or a restricted return line, a coolant temperature signal reading colder than the engine actually is so the ECM keeps enriching, or a mass air flow signal reading high. Replacing a sensor that is telling the truth leaves the customer with the same complaint.

Q47. Answer: C

Pedal pulsation: rotor thickness variation (DTV). Pulsation is a vibration felt through the pedal during braking. A rotor that is not perfectly uniform in thickness around its circumference pushes the caliper pistons in and out as it rotates — this pressure variation is felt as a pulsation through the pedal. DTV is caused by excessive runout (rotor wobble) that lets the pads touch and wear the rotor unevenly, or by improper wheel torque (over-torquing warps rotors). Check: measure rotor thickness at 8+ points.

Q48. Answer: D

Excessive travel with a pedal that is firm and holds is a mechanical clearance problem, not a hydraulic one. Drum brakes need the shoes held close to the drum, and the automatic adjuster takes up lining wear a little at a time so the pedal stays high. When the adjuster seizes with rust, or its lever is bent, unhooked or reinstalled wrongly, the shoes stay where the last adjustment left them and the gap grows as the linings wear, so more and more pedal travel is spent moving the shoes out to the drum before any braking starts. Each of the other answers produces a different pedal. A master cylinder bypassing internally gives a pedal that sinks slowly under steady pressure and firms up when pumped - excluded by the stem. Leaking wheel cylinders lose fluid, so the reservoir level falls and the linings and backing plate are wet - also excluded. Worn front disc pads add almost no travel, because the caliper piston follows the pad outward as the pad wears. Pull the drums, free and lubricate the adjuster, verify the lever action, adjust the shoes to the specified drag and then check pedal height and the parking brake adjustment.

Q49. Answer: D

Automatic transmission fluid is checked hot and running, because that is the condition the marks on the dipstick were calibrated for. Fluid expands as it warms, and a large part of the charge is not in the pan at all while the engine runs: it is out in the torque converter, the cooler, the lines and the apply circuits. Stop the engine and that fluid drains back into the pan, so a level taken with the engine off reads high and invites a technician to remove fluid the transmission needs. Check it cold and the fluid has not expanded, so the level reads low and invites overfilling - and an overfilled automatic aerates its fluid, which foams, loses pressure, and makes clutches slip and shift quality suffer. Bring the transmission to operating temperature, leave the engine idling on level ground, move the selector through the ranges and return it to the position the service information gives, then read the dipstick, wipe it and read again to confirm. Some units are specified differently again - a minority are read with the engine stopped immediately after a run, and many late units have no dipstick at all and are checked at a level plug with the fluid held inside a specified temperature window read on a scan tool - which is one more reason to read the procedure for the unit in front of you rather than assume. Level and condition are checked before any slipping or shift complaint is chased into the transmission.

Q50. Answer: B

Pressure at low demand proves only that the pump can reach pressure — it does not prove the system can supply the volume the engine needs at high demand. A restriction anywhere in the supply path (a collapsing pickup sock, a plugged filter, a kinked line, a tiring pump) still lets the pump build rated pressure at idle, because the engine is drawing very little fuel and the pump has time to make it up. Raise the demand and the pump cannot keep the rail full: pressure falls away, the mixture goes lean, and the driver feels the power loss. That is why a complaint that appears only under load has to be tested under load. Measure delivered volume over a timed interval into a graduated container to the manufacturer's procedure, or leave a gauge and scan tool connected and watch rail pressure and fuel trim while the engine is genuinely loaded. Repeating the same low-demand test, checking injector windings, or running a balance test at idle all sample the engine in the condition where it already works.

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